EXPERIMENT - 02

Aim -

To convert the ER diagram of an Indian e-commerce platform into a relational schema and implement it using MySQL with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and ON DELETE SET NULL constraints. To insert sample data and demonstrate referential integrity violations.

SOURCE CODE

```
CREATE DATABASE EcommerceDB;
USE EcommerceDB;
CREATE TABLE Customer (
Customer_ID INT PRIMARY KEY,
first_Name VARCHAR(30) NOT NULL,
Middle_Name VARCHAR(30),
Last_Name VARCHAR(30) NOT NULL,
Email VARCHAR(100) NOT NULL UNIQUE,
Phone VARCHAR(15) NOT NULL UNIQUE,
House VARCHAR(50),
Pin VARCHAR(10),
City VARCHAR(50).
);
CREATE TABLE Seller (
 Seller_ID INT PRIMARY KEY,
Seller_Name VARCHAR(100) NOT NULL,
Email VARCHAR(100) NOT NULL UNIQUE
Phone VARCHAR(15) NOT NULL UNIQUE
);
```

```
CREATE TABLE Category(
    Category_ID INT PRIMARY KEY,
    Category_Name VARCHAR(50) NOT NULL UNIQUE,
    Description VARCHAR(200)
);

CREATE TABLE Product(
    Product_ID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Price DECIMAL(10.2) NOT NULL,
    Stock INT NOT NULL,
    Category_ID INT,
    Seller_ID INT,

    FOREIGN KEY (Category_ID)
        REFERENCES Category(Category_ID)
        ON DELETE SET NULL,
    FOREIGN KEY (Seller_ID)
        REFERENCES Seller(Seller_ID)
        ON DELETE SET NULL
    );

    CREATE TABLE Orders(
        Order_ID INT PRIMARY KEY,
        Customer_ID INT NOT NULL,
        Date DATE NOT NULL,
        Amount DECIMAL(10,2) NOT NULL,
    FOREIGN KEY (customer_ID)
        REFERENCES Customer(customer_ID)
        ON DELETE CASCADE
        );
```

```
CREATE TABLE OrderITEM (
  Order_ID INT,
  Product_ID INT,
PRIMARY KEY (order_ID, Product-ID),
 FOREIGN KEY (order_ID)
  REFERENTIAL ORDER (order_ID)
  ON DELETE CASCADE,
FOREIGN KEY (Product_ID)
  REFERENCES Product (Product_ID)
     ON DELETE CASCADE
   );
  CREATE TABLE PAYMENT (
    Payment_ID INT PRIMARY KEY,
   Order_ID INT NOT NULL UNIQUE,
   Amount DECIMAL (10,2) NOT NULL,
   Method VARCHAR (30) NOT NULL,
    Status VARCHAR (30) NOT NULL,
      FOREIGN KEY (Order_ID)
       REFERENCES Orders (Order_ID)
        ON DELETE CASCADE
     );
CREATE TABLE Delivery (
  Delivery_ID INT PRIMARY KEY,
  Order_ID INT NOT NULL UNIQUE,
    Delivery_Date DATE,
  Status VARCHAR (30) NOT NULL,
```

```
    Tracking VARCHAR (50) UNIQUE,
    FOREIGN KEY (order_ID)
        REFERENCES Orders (order_ID)
         ON DELETE CASCADE
         );
```

INSERT SAMPLE DATA

Customer

```
INSERT INTO customerss
 VALUES
(1, 'Shreya', 'shrey216@gmail.com', '6390188xxxx', '12A',
'110001', 'Kanpur'),
(2, 'Vaishnavi', 'srinastanavaishnavi2405@gmail.com',
'6393789128', '12A', '211008', 'Prayagraj');
```

Seller

```
 INSERT INTO Seller
 VALUES
(101, 'Tech World', 'tech@gmail.com', '9811111111');
(102, 'Fashion hub', 'fashion@gmail.com', 8822222222);
```

Product

```
INSERT INTO Product
 VALUES
(301, 'LAPTOP', 65000.00), 10, 201, 101);
(302, 'Smartphone', '30000.00, 20, 201, 100);
(303, 'Jacket', '2500.00', 15, 202, 102);
```

Orders

```
INSERT INTO orders
 VALUES
(401,1,'2026-08-19', 65000.00),
(402, 2, '2026-08-19', 30000.00);
```

Order Item

```
INSERT INTO OrderItem
 VALUES
 (401,301);
 (402,302);
```

Payment

```
INSERT INTO Payment
 VALUES
 (501,401,65000.00,'UPI', 'Successful');
 (502,402,30000.00,'Card', 'Successful');
```

Delivery

```
INSERT INTO Delivery
 VALUES
 (601,401,'2026-08-22', 'In Transit', 'TRK10001')
 (602,402,'2026-08-22', 'Processing', 'TRK20002')
```

DEMONSTRATION OF REFERENTIAL INTEGRITY VIOLATION:

1) FOREIGN KEY VIOLATION

Trying to insert an order for a customer who does not exist:

```
INSERT INTO Orders
VALUES
(403, 999, '2026-08-19', 5000.00);
```

Output

```
ERROR 1452 (23000);
Cannot add or update a child row:
a foreign key constraint fails
```

Here, customer_ID = 999 does not exist in the customers table.

2. Primary key violation

```
INSERT INTO Customers
VALUES
(1, 'Rahul', 'rahul@gmail.com', '6393782163', '20A',
'110002', 'Delhi');
```

Output

```
ERROR 1062 (23000);
Duplicate entry '1' for key 'PRIMARY' Because
customer_ID = 1 already exists.
```

UNIQUE Constraint Violation -

```
INSERT INTO customers
VALUES
(3, 'Rahul', 'rahul@gmail.com', '6393782163', '20A',
```

Output

ERROR 1062 (23000):
Duplicate entry 'Pari@gmail.com' for key 'Email'
Because the Email address is declared as UNIQUE

DEMONSTRATION OF ON-DELETE CASCADE

First, delete customer1:

```
DELETE FROM Customers
WHERE Customer_ID = 1;
```

Because orders Customer_ID has ON DELETE CASCADE all orders belonging to customer 1 will also be deleted.

The related Order Item, Payment and Delivery records will also be deleted through their cascading foreign keys.

DEMONSTRATION of ON DELETE SET NULL.

Delete a Seller:

```
DELETE FROM seller
WHERGE seller_ID = 101;
```

Since Product.Seller_ID uses ON DELETE SET NULL the seller information will be removed but the products will remain.

Check the Product table:

```
SELECT Product_ID, Name, seller_ID
FROM Product;
```

Output

Product_ID	Name	Seller_ID
301	Laptop	Null
302	Smartphone	NULL
303	Jacket	102

RELATIONAL SCHEMA

Customer (
Customer_ID PK,
Name,
Email UNIQUE,
Phone UNIQUE,
House,
Pin,
City
)

Seller (
Seller_ID PK,
Seller_Name,
Email UNIQUE,
Phone UNIQUE,
)

Category (
Category_ID PK,
Category_Name UNIQUE,
Description
)

Product (
Product_ID PK,
Name,
Price,
stock,
Category_ID PK
Seller_ID FK
)

Orders (
Order_ID PK,
Customer_ID FK,
Date,
Amount
)

Order Item (
Order_ID PK, FK,
Product_ID PK, FK
)

Payment (
Payment_ID PK,
Order_ID FK UNIQUE,
Amount,
Method,
Status
)

Delivery (
Delivery_ID PK,
Order_ID FK UNIQUE,
Delivery_Date,
Status,
Tracking UNIQUE
)

OUTPUT:

The given ER diagram is successfully converted into a relational schema. All tables are created successfully with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and ON DELETE SET NULL constraints. Sample data is inserted successfully and referential integrity violations are demonstrated.

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